

NovaChain Logistics Inc., USA

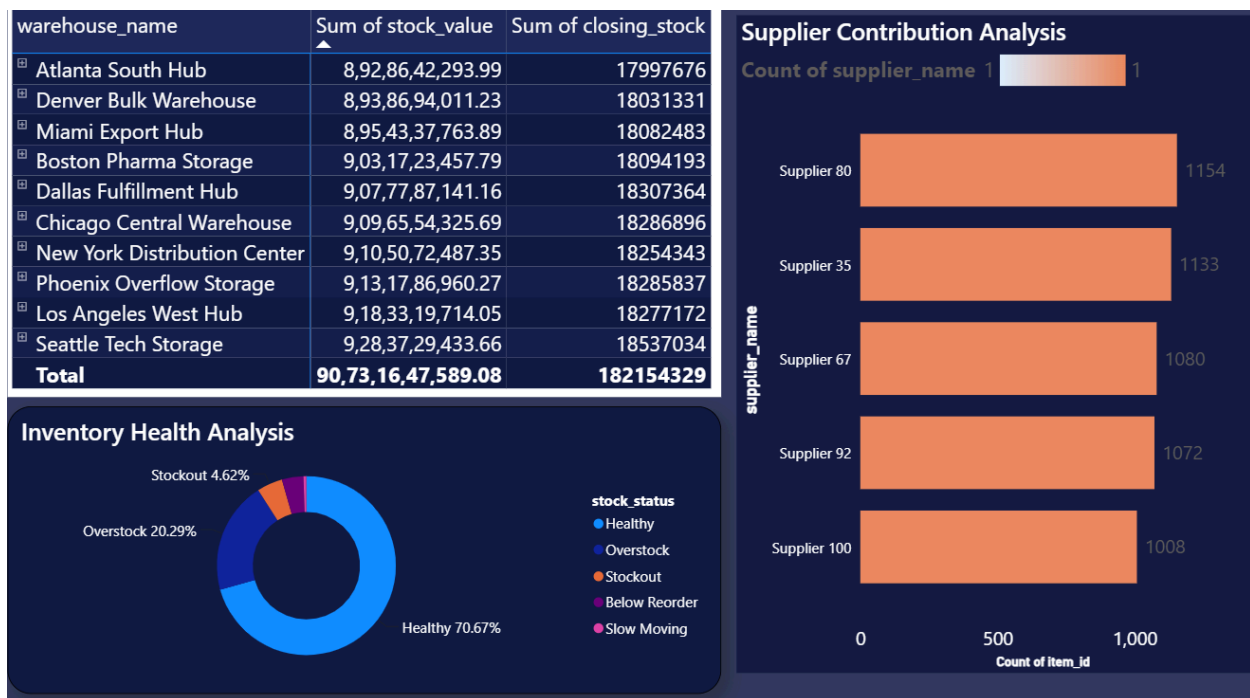
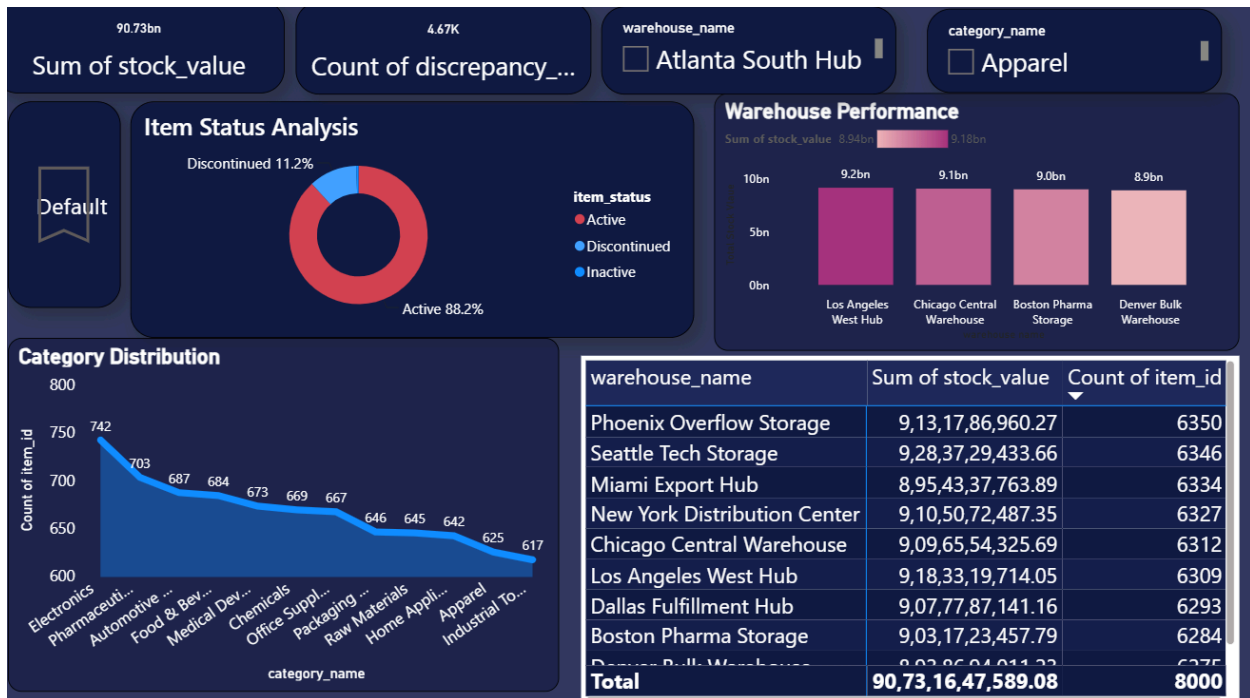
mail1 :Requirement for Basic Inventory Overview Dashboard in Power BI

Business Summary:

Provides a complete overview of inventory stock, item status, warehouse performance, supplier contribution, and inventory health.

- Manages 8,000 inventory items with a total stock value of 90.73bn.
- 88.2% of items are Active, while 11.2% are Discontinued, indicating a healthy product portfolio.
- Electronics is the largest product category with the highest number of inventory items.
- Los Angeles West Hub, Chicago Central Warehouse, and Boston Pharma Storage hold the highest inventory stock value.
- Supplier analysis identifies the top suppliers contributing the most inventory across warehouses.
- Inventory health analysis shows that most inventory is healthy, with a smaller percentage of overstock, stockout, and below-reorder items.
- The dashboard helps optimize inventory planning, warehouse performance, supplier management, and stock availability, supporting better business decisions.

Dashboard Screenshot:



Mail 2 – Inventory Movement & Stock Analytics

1 Monthly Inventory Movement Trend

Total MOVEMENT :

```
total movement = sum(  
    'public inventory_movement_log'[quantity]  
)
```

High movement month:

Highest Movement Month =

```
CALCULATE(  
    MAX('public inventory_movement_log'[Month]),  
    TOPN(  
        1,  
        SUMMARIZE(  
            'public inventory_movement_log',  
            'public inventory_movement_log'[Month],  
            "TotalMovement", SUM('public inventory_movement_log'[quantity])  
        ),  
        [TotalMovement], DESC  
    )  
)
```

2 Warehouse Movement Performance

Total Movement Value =

```
SUMX(  
    'public inventory_movement_log',  
    'public inventory_movement_log'[quantity] *  
    'public inventory_movement_log'[unit_cost]  
)
```

Movement Frequency

Movement Frequency =

```
COUNT('public inventory_movement_log'[movement_id])
```

5 Stockout Analysis

Total stockout Value:

Total Stockout =

SUM('public inventory_kpi_summary'[stockout_count]))

6 Inventory Turnover KPI

Average Stock :

Average Stock Movement =

AVERAGE('public inventory_movement_log'[quantity])

Inventory Turnover Ratio:

Inventory Turnover Ratio =

SUM('public inventory_kpi_summary'[inventory_turnover])

7 Inventory Value Movement Analysis

Stock Value Moved:

Stock Value Moved1 =

**SUMX(
 'public inventory_movement_log',
 'public inventory_movement_log'[quantity] *
 RELATED('public inventory_item_master'[unit_cost])
)**

Quantity Moved:

Quantity Moved =

SUM('public inventory_movement_log'[quantity])

Business Summary:

The inventory dashboard provides a comprehensive overview of warehouse operations, stock movement, inventory turnover, and stock availability across multiple warehouses and product categories.

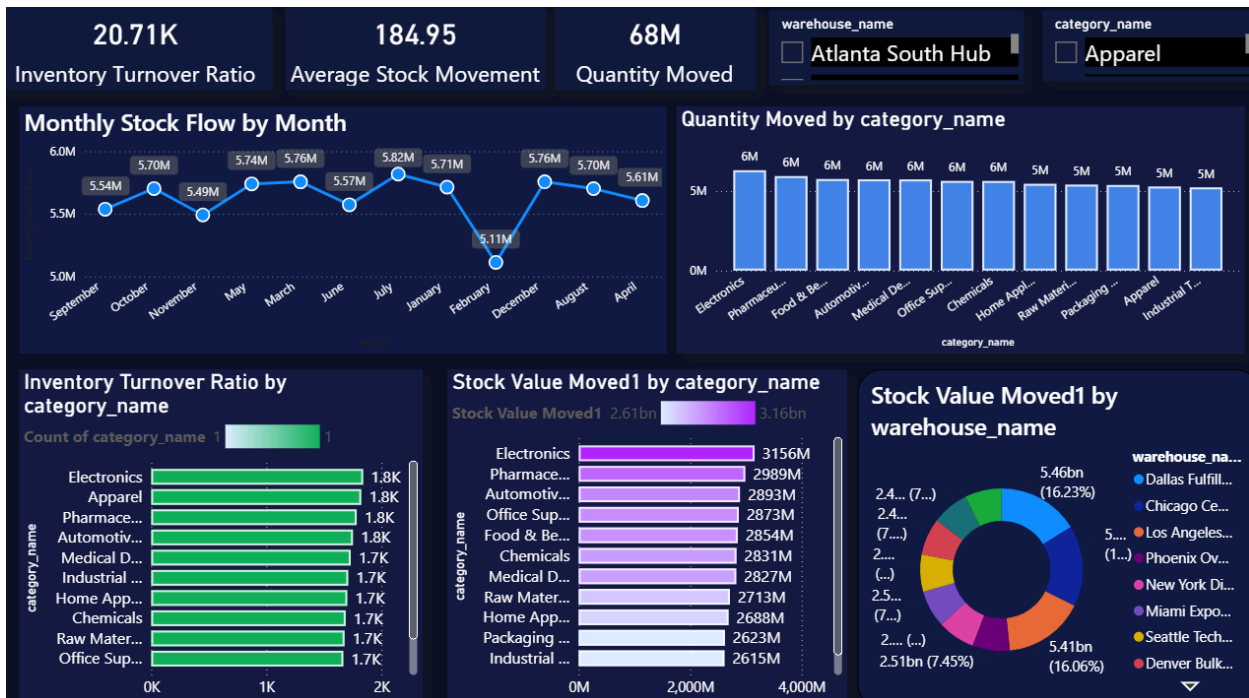
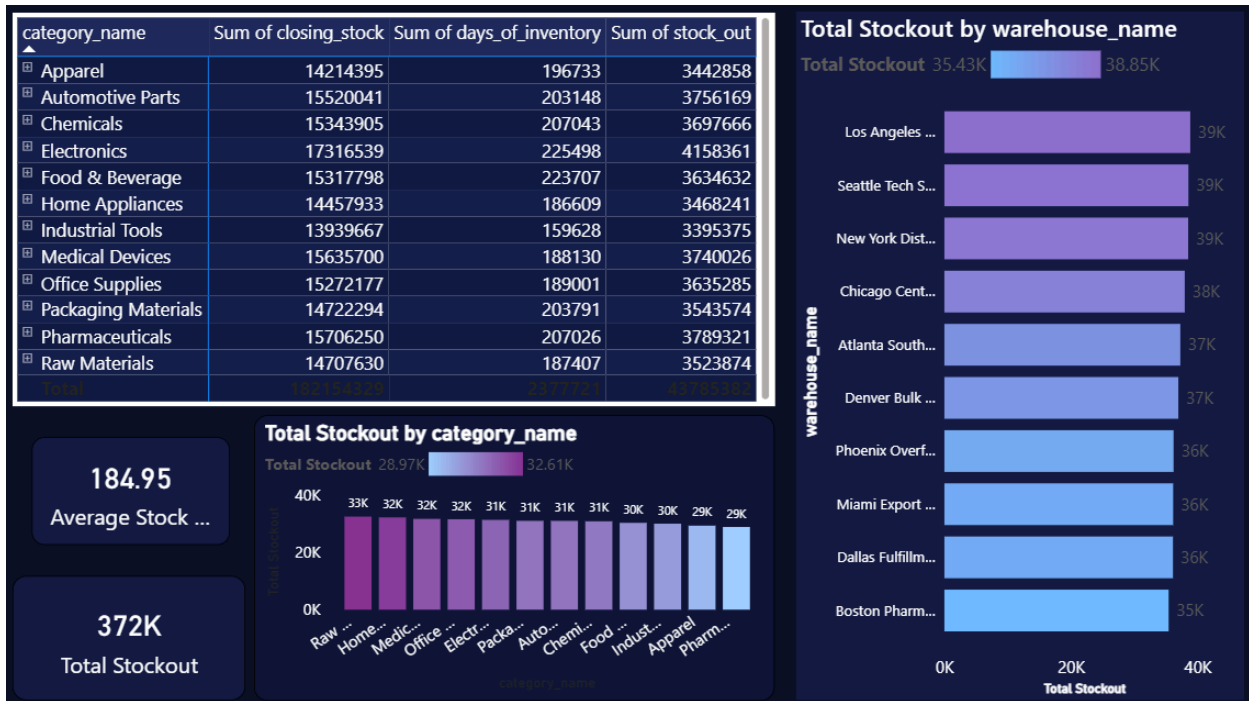
Key Business Highlights

- Total Quantity Moved: 68 Million units, indicating high inventory movement across all warehouses.
- Inventory Turnover Ratio: 20.71K, showing strong inventory circulation and efficient stock utilization.

- Average Stock Movement: 184.95 units, reflecting steady product movement throughout the business.
- Total Stock Units: 474 Million units currently available across all warehouses.
- Movement Frequency: 365K inventory transactions were recorded.

Dashboard Screenshot





warehouse_name	Below Reorder	Healthy	Overstock	Total
Atlanta South Hub	58992	12449798	5401964	17910754
Boston Pharma Storage	57142	12498827	5432748	17988717
Chicago Central Warehouse	61009	12624603	5492472	18178084
Dallas Fulfillment Hub	54875	12589409	5567363	18211647
Denver Bulk Warehouse	60049	12306535	5549943	17916527
Los Angeles West Hub	53888	12631159	5488095	18173142
Miami Export Hub	56593	12439157	5501268	17997018
New York Distribution Center	53394	12554224	5529616	18137234
Phoenix Overflow Storage	54477	12633340	5456255	18144072
Seattle Tech Storage	49152	12905168	5482978	18437298
Total	559571	125632220	54902702	181094493

Mail 3 – Discrepancy, Validation & Task Productivity Analytics

4 Validation Issue Analysis

Open Issues:

Open Issues =

```
CALCULATE(
    COUNT('public inventory_validation_log'[validation_id]),
    'public inventory_validation_log'[validation_status] = "Open"
)
```

Closed Issues:

Closed Issues =

```
CALCULATE(
    COUNT('public inventory_validation_log'[validation_id]),
    'public inventory_validation_log'[validation_status] = "Closed"
)
```

High Severity Issues:

High Severity Issues =

```
CALCULATE(
    COUNT('public inventory_validation_log'[validation_id]),
```

```
'public inventory_validation_log'[severity] = "High"  
)
```

6 Staff Productivity Analysis

Discrepancy Qty:

```
Discrepancy Qty =  
VAR WH = SELECTEDVALUE('public warehouses'[warehouse_name])  
RETURN  
CALCULATE(  
    SUM('public vw_discrepancy_analysis'[discrepancy_quantity]),  
    'public vw_discrepancy_analysis'[warehouse_name] = WH  
)
```

8 Discrepancy vs Task Delay Correlation

Operational Risk:

```
Operational Risk =  
IF(  
    NOT ISINSCOPE('public warehouses'[warehouse_name]),  
    BLANK(),  
    VAR Delayed = [Delayed Tasks]  
    VAR Discrepancy = [Discrepancy Qty]  
    VAR Validation = [Validation Issues]  
    RETURN  
    SWITCH(  
        TRUE(),  
        Delayed >= 2650 &&  
        Discrepancy >= 50000 &&  
        Validation >= 5800, "🔴 High Risk",  
  
        Delayed >= 2580 &&  
        Discrepancy >= 15000 &&  
        Validation >= 5700, "🟡 Medium Risk",  
  
        "🟢 Low Risk"
```

```

)
)
Task Efficiency:
Task Efficiency % =
DIVIDE(
    SUM('public vw_task_productivity_summary'[completed_tasks]),
    SUM('public vw_task_productivity_summary'[total_tasks]),
    0
)

```

Severity Count:

```

Severity Count =
COUNTROWS('public inventory_discrepancy_log')

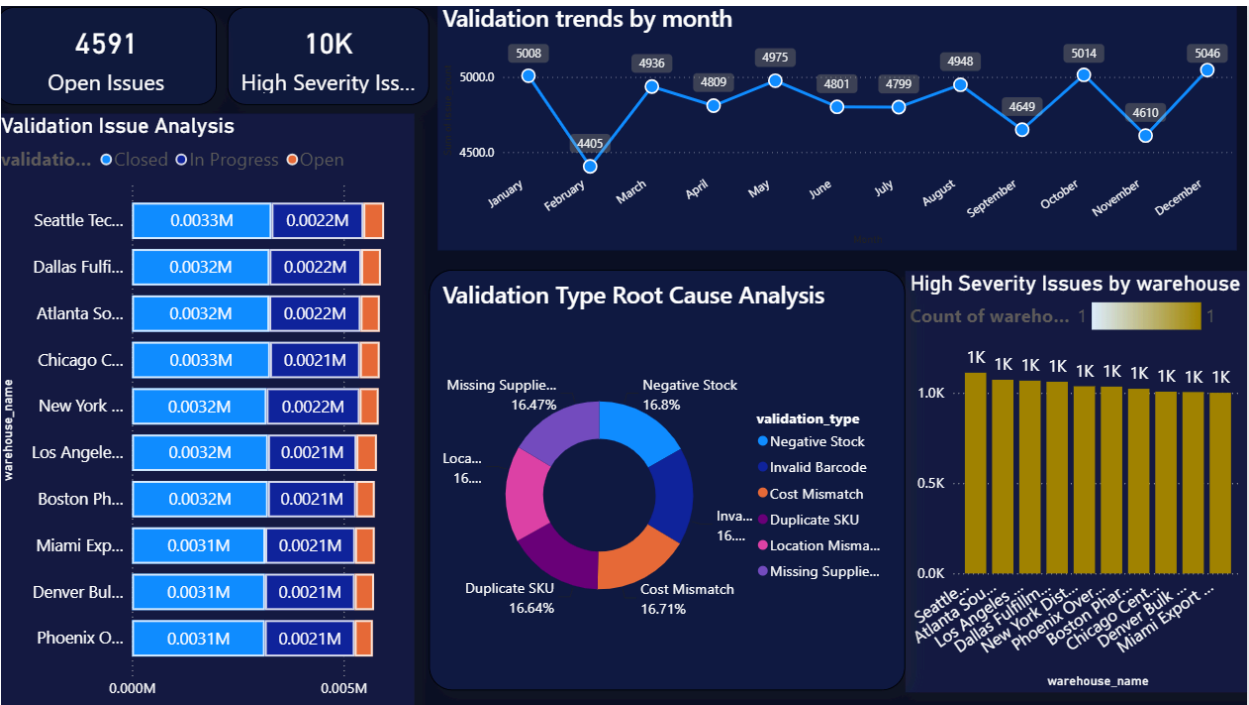
```

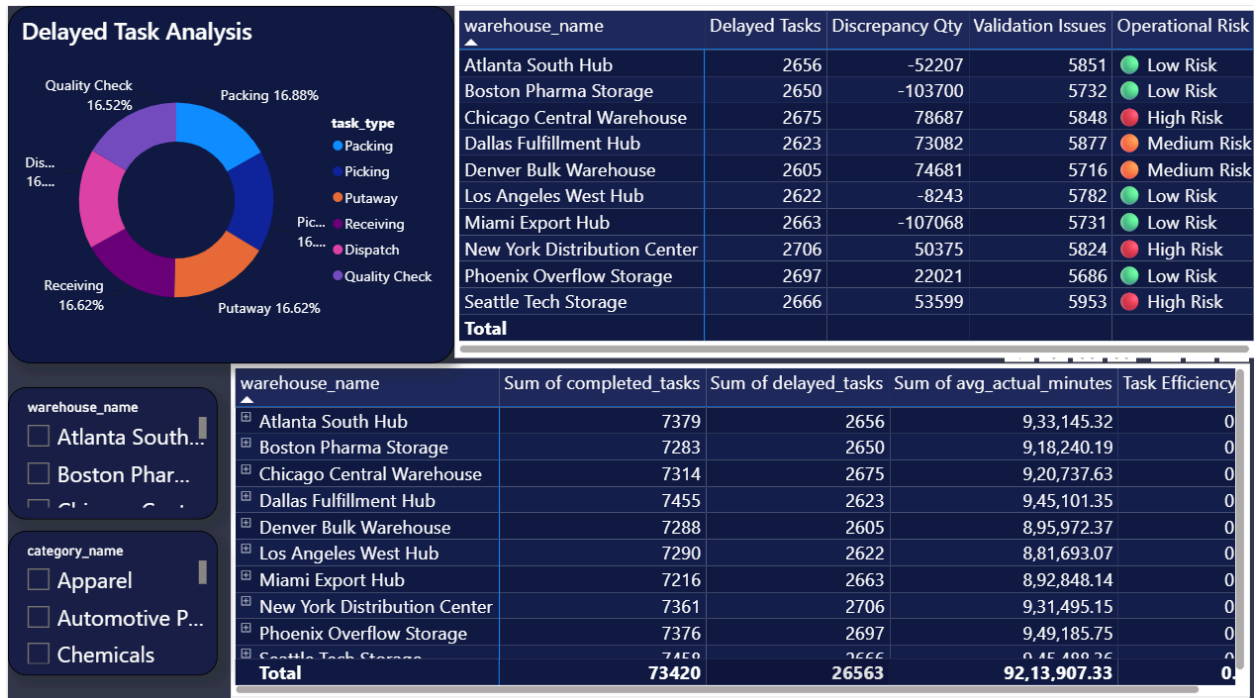
Business Insights:

- Provides a complete view of warehouse operations, inventory discrepancies, validation issues, and delayed task performance.
- Tracks 81K discrepancy quantity with a total discrepancy value of 15.78bn.
- 92.27% of discrepancies are classified as High Severity, requiring immediate attention.
- Monitors 4,591 open validation issues and 10K high-severity validation issues across warehouses.
- Identifies major issue causes such as Supplier Shortage, Barcode Scan Errors, Wrong Bin Location, Invalid Barcode, Cost Mismatch, and Negative Stock.
- Evaluates warehouse task performance by tracking completed tasks, delayed tasks, validation issues, and operational risk.
- Classifies warehouses into Low, Medium, and High Risk to support operational decision-making.
- Analyzes delays across Picking, Packing, Putaway, Receiving, Dispatch, and Quality Check processes.
- Enables warehouse managers to identify high-risk locations, reduce discrepancies, improve data quality, minimize task delays, and enhance overall

operational efficiency.

Dashboard Screenshots:





Mail 4 – Executive Supply Chain

High Stock Value but Low Movement Analysis

Dead Stock Value =

SUMX (

VALUES ('public item_categories'[category_name]),

IF ([Dead Stock Flag] = "High Value / Low Movement", [Total Stock Value])

)

Dead Stock Flag:

Dead Stock Flag =

VAR StockVal = [Total Stock Value]

VAR MoveVel = [Movement Velocity (90D)]

RETURN

IF (

StockVal > 50000 && MoveVel < 5,

"High Value / Low Movement",

```

        "Normal"
    )

-- Dead Stock % of Total Inventory
    Dead Stock % of Total Inventory =
    DIVIDE ( [Dead Stock Value], [Total Stock Value] )

```

4:Discrepancy + Validation Risk Analysis

```

Operational Risk Score =
0.5 * SUM('public inventory_discrepancy_log'[discrepancy_value])
+
0.3 * COUNT('public inventory_validation_log'[validation_id])
+
0.2 * CALCULATE(
    COUNT('public task_allocation_log'[task_id]),
    'public task_allocation_log'[task_status] = "Delayed"
)

```

Operational Risk:

```

Operational Risk =
VAR RankRisk =
    RANKX(
        ALL('public warehouses'[warehouse_name]),
        [Operational Risk Score],
        ,
        DESC,
        DENSE
    )

```

```

RETURN
SWITCH(
    TRUE(),
    RankRisk <= 3, " High Risk",

```

```

RankRisk <= 7, " Medium Risk",
" Low Risk"
)

```

5:Supplier Risk Analysis

```

Supplier Risk Score =
0.5 * SUM('public inventory_discrepancy_log'[discrepancy_value])
+
0.3 * COUNT('public inventory_validation_log'[validation_id])
+
0.2 * SUM('public inventory_kpi_summary'[slow_moving_count])

```

6 :Fill Rate & Inventory Optimization Analysis

```

Fill Rate % =
DIVIDE(
    SUM('public inventory_profiling'[stock_out]),
    SUM('public inventory_profiling'[stock_in]),
    0
)

```

--Inventory Turnover

```

Inventory Turnover =
DIVIDE(
    SUM('public inventory_profiling'[stock_out]),
    AVERAGE('public inventory_profiling'[closing_stock]),
    0
)

```

--Overstock Items

Overstock Items =

```
CALCULATE(
  COUNTROWS('public inventory_profiling'),
  FILTER(
    'public inventory_profiling',
    'public inventory_profiling'[closing_stock] >
    RELATED('public inventory_item_master'[max_stock_level])
  )
)
```

--Slow Moving Inventory

Slow Moving Inventory =

```
CALCULATE(
  COUNTROWS('public inventory_profiling'),
  'public inventory_profiling'[days_of_inventory] > 90
)
```

--Stockout Frequency

Stockout Frequency =

```
CALCULATE(
  COUNTROWS('public inventory_profiling'),
  'public inventory_profiling'[closing_stock] = 0
)
```

7:Overstock & Slow-Moving Inventory Analysis

Inventory Optimization Indicator =

VAR AvgOverstock =

```
AVERAGEX(
  ALL('public item_categories'),
  [Overstock Items]
)
```

```

VAR AvgSlowMoving =
AVERAGEX(
    ALL('public item_categories'),
    [Slow Moving Inventory]
)

VAR AvgTurnover =
AVERAGEX(
    ALL('public item_categories'),
    [Inventory Turnover]
)

VAR Score =
IF([Overstock Items] > AvgOverstock,1,0) +
IF([Slow Moving Inventory] > AvgSlowMoving,1,0) +
IF([Inventory Turnover] < AvgTurnover,1,0)

RETURN
SWITCH(
    TRUE(),
    Score = 3," High Priority",
    Score = 2," Medium Priority",
    Score = 1," Low Priority",
    "Healthy"
)

```

8:Executive Attention Dashboard

```

Total Stock Value =
SUM ( 'public inventory_kpi_summary'[total_stock_value] )

High-Risk Category Flag (Column) =
VAR StockVal = CALCULATE ( SUM ( 'public

```

```

inventory_kpi_summary'[total_stock_value] ) )
VAR MaxDate = CALCULATE ( MAX ( 'public
inventory_movement_log'[movement_date] ), ALL ( 'public
inventory_movement_log' ) )
VAR MoveVelocity =
CALCULATE (
SUM ( 'public inventory_movement_log'[quantity] ),
'public inventory_movement_log'[movement_type] = "Outbound",
'public inventory_movement_log'[movement_date] >= MaxDate - 90
)
RETURN
IF ( StockVal > 50000 && MoveVelocity < 5, "High-Risk Category", "Normal" )

```

```

Operational Bottleneck Count =
CALCULATE (
COUNTROWS ( 'public task_allocation_log' ),
'public task_allocation_log'[actual_duration_minutes] > 'public
task_allocation_log'[expected_duration_minutes]
)

```

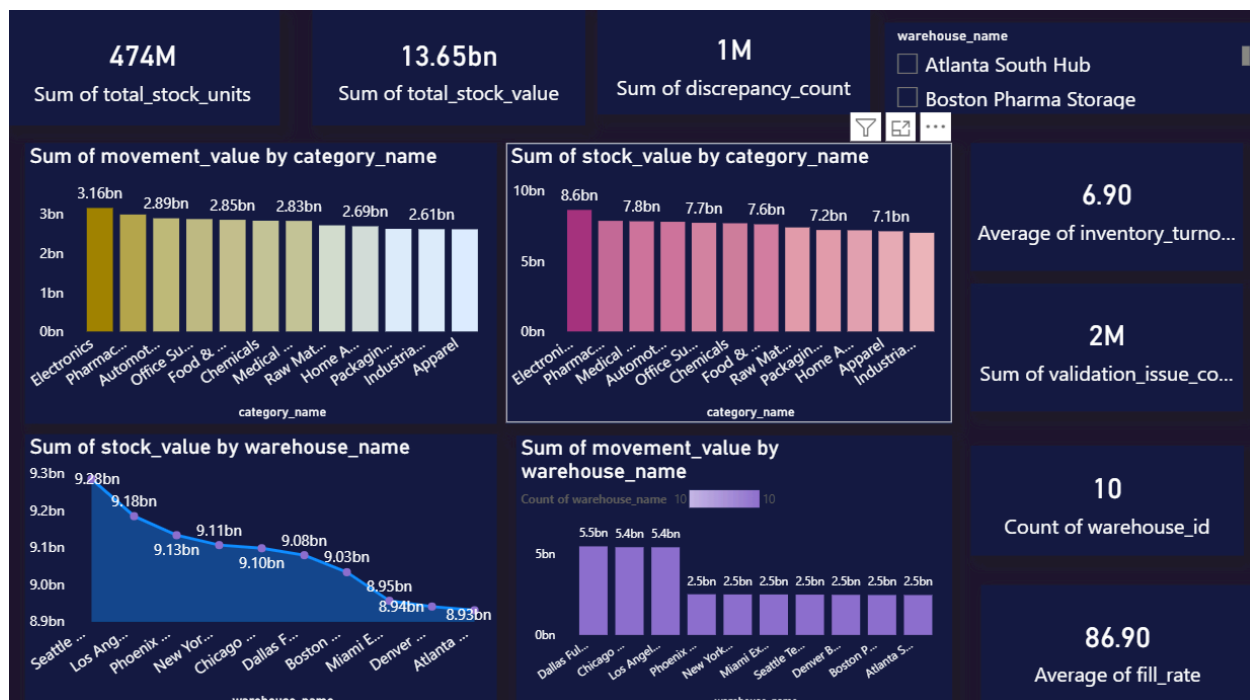
Business Insights:

Provides a comprehensive view of inventory optimization, warehouse performance, supplier risk, and operational efficiency.

- Manages 474M total stock units with a total stock value of 13.65bn across 10 warehouses.
- Tracks 58K validation issues, 1M inventory discrepancies, and a total discrepancy value of 15.78bn.
- Maintains an average Fill Rate of 86.90% and an Inventory Turnover of 30.05K, indicating stable inventory performance.
- Identifies 3,645 slow-moving items, 25K overstock items, and 5,777 stockout occurrences, highlighting optimization opportunities.

- Evaluates operational risk for each warehouse to prioritize corrective actions and improve performance.
- Classifies product categories using Inventory Optimization Indicators (Healthy, High Priority, Medium Priority, and Low Priority) for better inventory planning.
- Monitors supplier risk levels (Low, Medium, and High Risk) to strengthen supplier performance and reduce supply chain risks.
- Supports better decision-making by improving inventory turnover, reducing overstock and stockouts, enhancing supplier management, and increasing warehouse operational efficiency.

Dashboard Screenshots:





3645
Slow Moving Items

30.05K
Inventory Turnover

25K
Overstock Items

5777
Stockout Frequency

Count of supplier_id by severity

severity	Count	Percentage
High	150	33.33%
Low	150	33.33%
Medium	150	33.33%

Fill Rate % by warehouse_name

warehouse_name	Fill Rate %
Boston Pharma S...	100
Chicago Central ...	100
Phoenix Overflo...	100
Los Angeles Wes...	100
Atlanta South Hub	100
Dallas Fulfillment...	100
New York Distrib...	100
Miami Export Hub	100
Denver Bulk War...	100
Seattle Tech Stor...	100

supplier_name

Supplier Risk Indica

Supplier 1	Low Risk
Supplier 10	Medium Risk
Supplier 100	Medium Risk
Supplier 101	Medium Risk
Supplier 102	Medium Risk
Supplier 103	Medium Risk
Supplier 104	Low Risk
Supplier 105	Medium Risk
Supplier 106	Medium Risk
Supplier 107	Medium Risk
Supplier 108	Medium Risk
Supplier 109	Medium Risk
Supplier 11	Medium Risk
Supplier 110	Low Risk
Supplier 111	Medium Risk
Supplier 112	Low Risk
Supplier 113	Medium Risk
Supplier 114	Medium Risk
Supplier 115	Medium Risk
Supplier 116	Medium Risk
Supplier 117	Medium Risk
Supplier 118	Medium Risk
Total	Medium Risk

